Semantic Analysis of Domain Data for Proactive Identification of Cybersecurity Indicators

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Combined EDA: PhishTank & Tranco

Exploratory data analysis on two datasets used in our phishing-detection pipeline:

1. **PhishTank** — verified phishing URLs (`../data/raw_input/verified_online.csv`)
2. **Tranco Top 1 M** — legitimate, popularity-ranked domains (`../data/raw_input/top-1m.csv`)

Relevance Statement

Our project synthesizes two datasets. First, we selected PhishTank, a dataset of user-reported phishing and spam URLs, verified by Cisco Talos, with metadata curated and compiled by the dataset hosts. Additionally, we use Tranco, which aggregates the top 1 million most visited domains on the day we downloaded it. The union of these two datasets minus their intersection serves as our baseline of sites — most of which are legitimate or at the very least not flagged as malicious — while the remainder are known phishing URLs. This pairing directly mirrors the approach taken in our literature review, where papers such as Saidani et al. and Sonowal & Gunikhan rely on contrasting verified malicious samples against high-traffic legitimate baselines to train semantic classifiers. The primary strength of these two datasets, and consequently our combined dataset, is their integrity: each record is complete with no null attributes. Our wrangling and preprocessing work largely consisted of organizing and decomposing URL components into discrete features for visualization and future ML analysis.

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from urllib.parse import urlparse
```

We have a requirement for this assignment that it run in Google Colab. As such, we pull the repo style and data from Github first.

In [2]:
```
import os

# Clone the repo (skip if already present)
if not os.path.exists('DS_ML_Project_ColabIntegration'):
    !git clone https://github.com/ns15468-gasou/DS_ML_Project_ColabIntegrati

os.chdir('DS_ML_Project_ColabIntegration/Project/notebooks')
print("Working directory:", os.getcwd())
```

```
Working directory: /home/nic/DS_ML_Project_ColabIntegration/Project/notebook
s/DS_ML_Project_ColabIntegration/Project/notebooks
```

PhishTank — Verified Phishing URLs

This section analyzes the PhishTank *verified-online* feed, a curated list of confirmed phishing URLs.
The dataset captures full URLs (scheme + host + path) along with the targeted brand and current online/offline status. [11] **Date accessed:** 03/05/2026

In [3]:
```
phish = pd.read_csv('../data/raw_input/verified_online.csv')
phish['target'] = phish['target'].str.replace('&amp;', '&', regex=False)
phish.head()
```

Out[3]:

	phish_id	url	phish_detail_url	
0	9358496	https://sgpass-fr033.com/	http://www.phishtank.com/phish_detail.php?phis...	2026-03-0
1	9358495	https://sgpass-fr033.com/pages/asset.php	http://www.phishtank.com/phish_detail.php?phis...	2026-03-0
2	9358485	https://dervi-emprend-ec.webcindario.com/	http://www.phishtank.com/phish_detail.php?phis...	2026-03-0
3	9358483	https://allegro.pl-oferta1081410.sbs	http://www.phishtank.com/phish_detail.php?phis...	2026-03-0
4	9358482	http://allegrolokalnie.oferta362639327346393.sbs	http://www.phishtank.com/phish_detail.php?phis...	2026-03-0

In [4]:
```
phish.shape
```

Out[4]:
```
(56045, 8)
```

In [5]:
```
phish['submission_time'] = pd.to_datetime(phish['submission_time'])
phish['verification_time'] = pd.to_datetime(phish['verification_time'])
phish.dtypes
```

Out[5]:
```
phish_id                           int64
url                                  str
phish_detail_url                     str
submission_time      datetime64[us, UTC]
verified                             str
verification_time    datetime64[us, UTC]
online                               str
target                               str
dtype: object
```

Null / Missing Value Analysis

In [6]:
```
phish.isnull().sum()
```

Out[6]:
```
phish_id             0
url                  0
phish_detail_url     0
submission_time      0
verified             0
verification_time    0
online               0
target               0
dtype: int64
```

In [7]:
```
(phish == '').sum()
```

Out[7]:
```
phish_id             0
url                  0
phish_detail_url     0
submission_time      0
verified             0
verification_time    0
online               0
target               0
dtype: int64
```

Target Brand Distribution

In [8]:
```
target_counts = phish['target'].value_counts().head(15)
print(target_counts)
```

```
target
Other                                   52773
Internal Revenue Service                  814
Allegro                                   654
Facebook                                  319
Microsoft                                 185
Optus                                     147
DHL                                       102
AT&T                                       77
Sumitomo Mitsui Banking Corporation        75
Amazon.com                                 67
Bradesco                                   62
Netflix                                    61
PayPal                                     57
Adobe                                      47
eBay, Inc.                                 41
Name: count, dtype: int64
```

In [9]:
```
plt.figure(figsize=(12,6))
sns.barplot(x=target_counts.values, y=target_counts.index)
plt.title('Figure 1: Top 15 Phishing Targets')
plt.xlabel('Count')
plt.ylabel('Target')
plt.tight_layout()
plt.show()
```

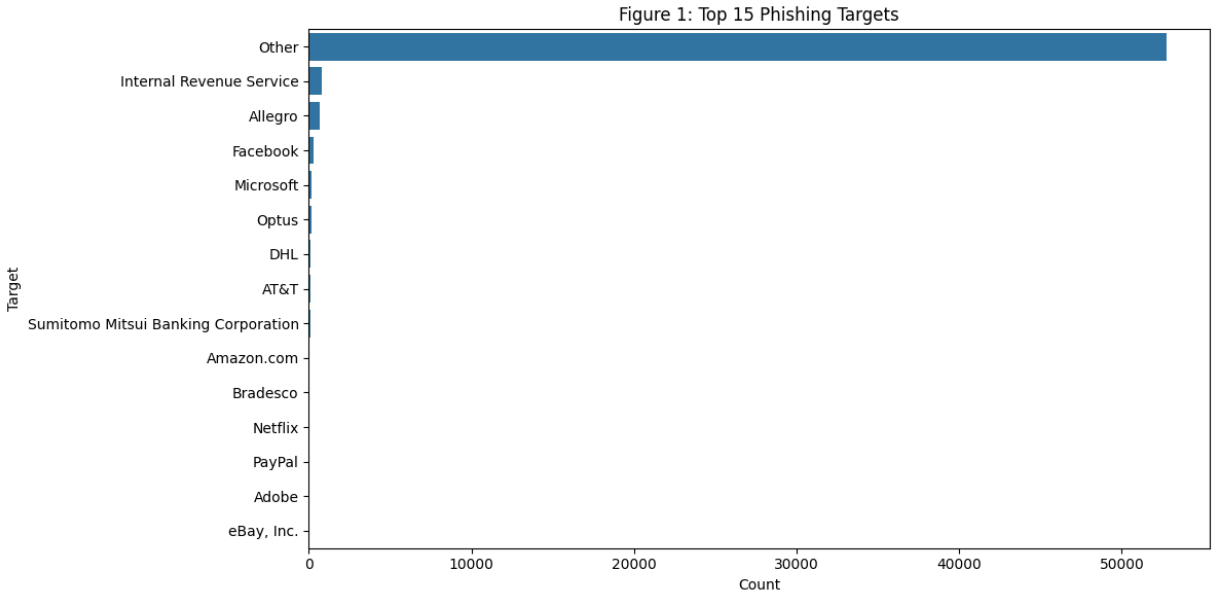


Figure 1 — Interpretation: The IRS dominates known phishing targets, followed by Allegro and Facebook. "Other" accounts for the vast majority of targets. "Other" is excluded in the below chart but note this represents a small fraction of the full dataset. "Other" representing the vast majority of data points suggests the opportunity to utilize unsupervised learning to categorize these data points based on common traits in their URLs.

In [10]:
```
target_counts_filtered = phish[phish['target'] != 'Other']['target'].value_c
plt.figure(figsize=(12,6))
sns.barplot(x=target_counts_filtered.values, y=target_counts_filtered.index)
plt.title('Figure 2: Top 15 Phishing Targets (excl. Other)')
plt.xlabel('Count')
plt.ylabel('Target')
plt.tight_layout()
plt.show()
```

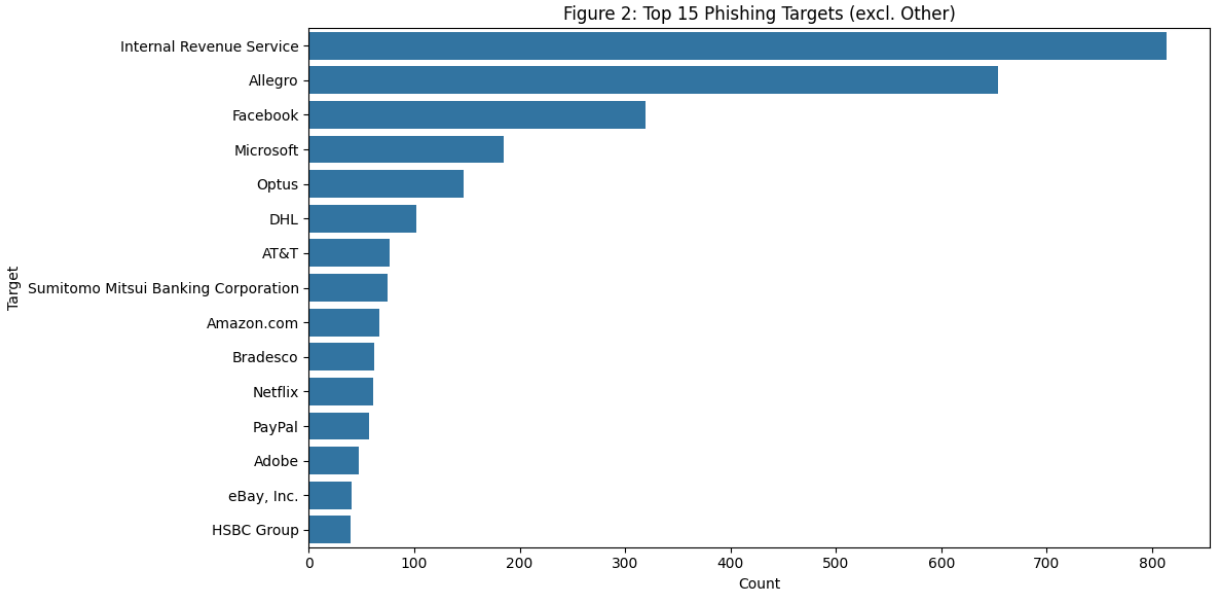


Figure 2 — Interpretation: Excluding 'Other', the IRS stands out as the most

impersonated brand by a wide margin, likely due to the high-stakes nature of tax-related fraud. Financial institutions and major tech platforms round out the top targets, consistent with findings in the literature on high-value phishing campaigns. Sites where it would not be extraordinary for users to input payment information are perhaps the most logical to masquerade as.

Online vs Offline Breakdown

In [11]:
```
online_counts = phish['online'].value_counts()
print(online_counts)

plt.figure(figsize=(6,4))
sns.barplot(x=online_counts.index, y=online_counts.values)
plt.title('Figure 3: Online vs Offline Phishing URLs')
plt.xlabel('Status')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

```
online
yes    56045
Name: count, dtype: int64
```

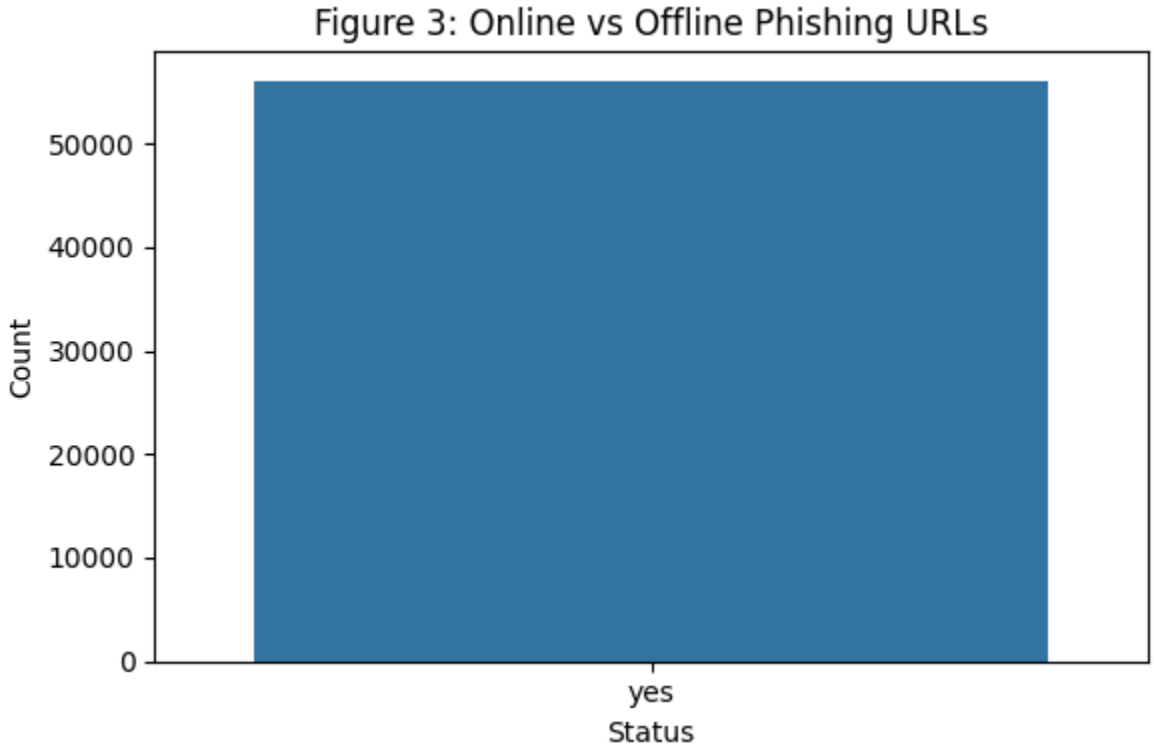


Figure 3 — Interpretation: This confirms the dataset represents active threats rather than historical ones, making it a strong signal for real-world phishing detection.

Dataset Limitations

This dataset is only the verified online feed, meaning this dataset is a subset of the greater PhishTank dataset. Future work could expand to use the full dataset. This dataset would be great for clustering/unsupervised ML work: the dominance of 'Other' in the target category shows the opportunity to clean/mine this data for meaningful categorization.

Data Dictionary — PhishTank

Feature	Data Type	Description	Example		---	---	---	---
phish_id	int	Unique identifier for each phishing submission	9358496
url	string	Full verified phishing URL	https://sgpass-fr033.com/
submission_time	datetime	Timestamp when URL was submitted to PhishTank	2026-03-03 19:50:49
verification_time	datetime	Timestamp when URL was verified as phishing	2026-03-03 20:12:24
online	string	Whether the phishing URL is currently active	yes		target	string	Brand or organization being impersonated	IRS
full_length	int	Total character length of the URL	47
host	string	Extracted hostname from the URL	sgpass-fr033.com
host_length	int	Character length of the hostname	16
path_depth	int	Number of path segments in the URL	2
dot_count	int	Number of dots in the full URL	3
hyphen_count	int	Number of hyphens in the full URL	2
digit_count	int	Number of digit characters in the URL	4
digit_ratio	float	Proportion of URL characters that are digits	0.085
subdomain_depth	int	Number of dots in the hostname (proxy for subdomain nesting)	1

URL Structure Analysis

Feature engineering and visualizations on URL structure — length, hosting domains, path depth, and character composition.

In [12]:
```
phish['host'] = phish['url'].apply(lambda x: urlparse(x).netloc.lower())
phish['full_length'] = phish['url'].str.len()
phish['host_length'] = phish['host'].str.len()
phish['path'] = phish['url'].apply(lambda x: urlparse(x).path)
phish['path_depth'] = phish['path'].apply(lambda p: p.strip('/').count('/')
phish['has_path'] = (phish['path'].str.strip('/') != '').astype(int)
phish['tld_ext'] = phish['host'].str.rsplit('.', n=1).str[-1]
phish['dot_count'] = phish['url'].str.count(r'\.')
phish['hyphen_count'] = phish['url'].str.count('-')
```

```
phish['digit_count'] = phish['url'].str.count(r'\d')
phish['digit_ratio'] = phish['digit_count'] / phish['full_length']
phish['subdomain_depth'] = phish['host'].str.count(r'\.')

print("PhishTank — Raw Features")
phish[['full_length', 'host_length', 'path_depth', 'dot_count', 'hyphen_coun
    'digit_count', 'digit_ratio', 'subdomain_depth']].describe()
```

```
PhishTank — Raw Features
```

Out[12]:

	full_length	host_length	path_depth	dot_count	hyphen_count	
count	56045.000000	56045.000000	56045.000000	56045.000000	56045.000000	56
mean	60.922866	19.275511	1.387457	1.974413	0.797127	
std	166.386061	12.977728	1.582218	1.025418	1.391185	
min	12.000000	4.000000	0.000000	1.000000	0.000000	
25%	23.000000	8.000000	0.000000	1.000000	0.000000	
50%	38.000000	16.000000	1.000000	2.000000	0.000000	
75%	62.000000	25.000000	2.000000	2.000000	1.000000	
max	25523.000000	147.000000	18.000000	67.000000	50.000000	3

In [13]:

```
clip_threshold = 0.98
phish_clipped = phish[['full_length', 'host_length', 'path_depth', 'dot_coun
                       'digit_count', 'digit_ratio', 'subdomain_depth']].app
    lambda col: col.clip(upper=col.quantile(clip_threshold)))
print("PhishTank — Features Clipped at 98th Percentile")
phish_clipped.describe()
```

```
PhishTank — Features Clipped at 98th Percentile
```

Out[13]:

	full_length	host_length	path_depth	dot_count	hyphen_count	
count	56045.000000	56045.000000	56045.000000	56045.000000	56045.000000	56
mean	57.060451	19.012561	1.378678	1.930181	0.739763	
std	47.987538	11.943231	1.547480	0.731294	1.030938	
min	12.000000	4.000000	0.000000	1.000000	0.000000	
25%	23.000000	8.000000	0.000000	1.000000	0.000000	
50%	38.000000	16.000000	1.000000	2.000000	0.000000	
75%	62.000000	25.000000	2.000000	2.000000	1.000000	
max	190.000000	60.000000	5.000000	4.000000	4.000000	

In [14]:

```
# URL length distribution (clipped to 98th percentile)
plt.figure(figsize=(10, 5))
sns.histplot(phish['full_length'], bins=80, edgecolor='white', alpha=0.85,
             binrange=(0, phish['full_length'].quantile(0.98)))
plt.axvline(phish['full_length'].mean(), ls='--', lw=1.2,
            label=f'Mean {phish["full_length"].mean():.1f}')
```

```
plt.title('Figure 4: PhishTank URL Length Distribution')
plt.xlabel('Character Length')
plt.ylabel('Frequency')
plt.tight_layout()
plt.legend()
plt.show()
```

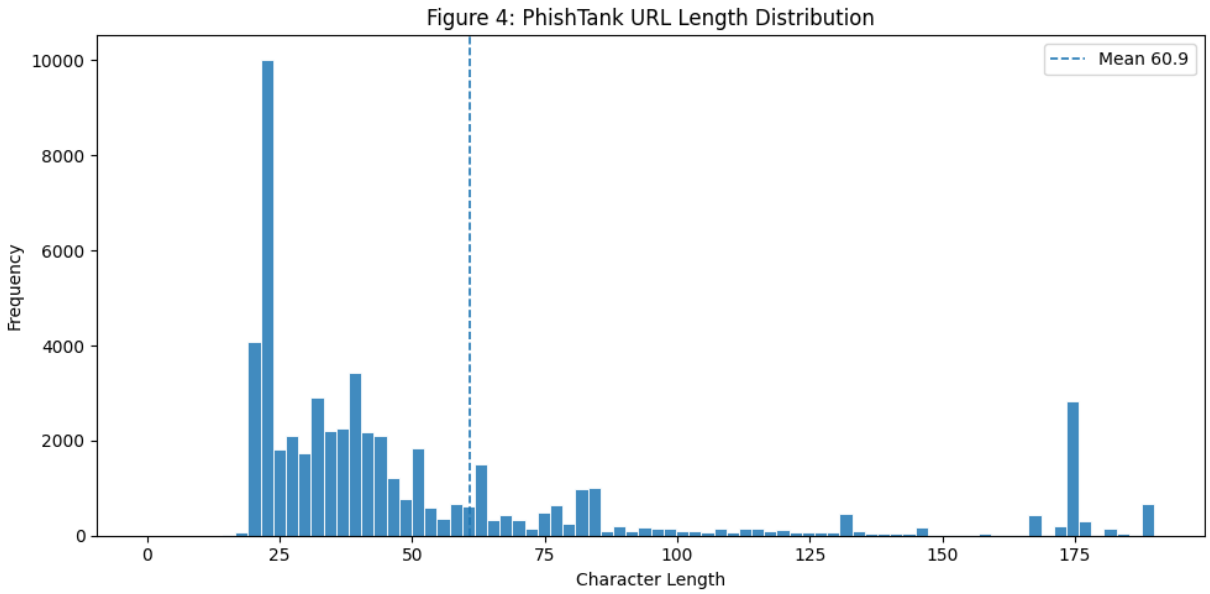


Figure 4 — Interpretation: PhishTank URL lengths are right-skewed with a long tail of very long URLs, clipped here at the 98th percentile for readability. The mean length is notably higher than Tranco domains (below), suggesting phishing URLs tend to be more verbose.

In [15]:
```
# Top 20 hosting domains
host_counts = phish['host'].value_counts().head(20)

fig, ax = plt.subplots(figsize=(10, 6))
host_counts.plot.barh(ax=ax, edgecolor='white')
ax.set_title('Figure 5: PhishTank — Top 20 Hosting Domains')
ax.set_xlabel('Number of Phishing URLs')
ax.invert_yaxis()

for i, v in enumerate(host_counts):
    ax.text(v + 1, i, str(v), va='center', fontsize=9)

plt.tight_layout()
plt.show()
```

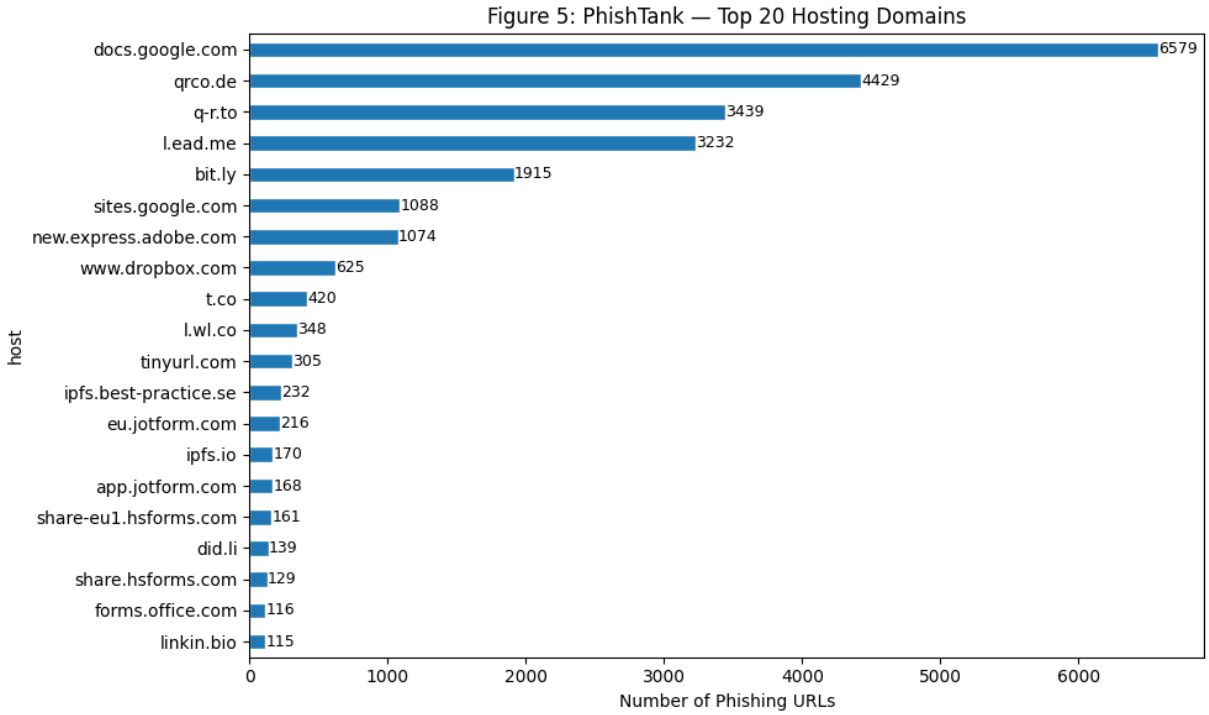


Figure 5 — Interpretation: A significant portion of phishing URLs are hosted on a small number of major platforms such as web hosting providers and cloud services. This indicates attackers frequently exploit the perceived legitimacy of established infrastructure, making domain-level detection alone insufficient.

In [16]:
```
# Path depth distribution (clipped to 98th percentile)
fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(phish['path_depth'],
        bins=range(0, int(phish['path_depth'].quantile(0.98)) + 2),
        edgecolor='white', alpha=0.85)
ax.set_title('Figure 6: PhishTank — URL Path Depth Distribution (≤98th pctl)
ax.set_xlabel('Path Depth (number of / segments)')
ax.set_ylabel('Frequency')
plt.tight_layout()
plt.show()
```

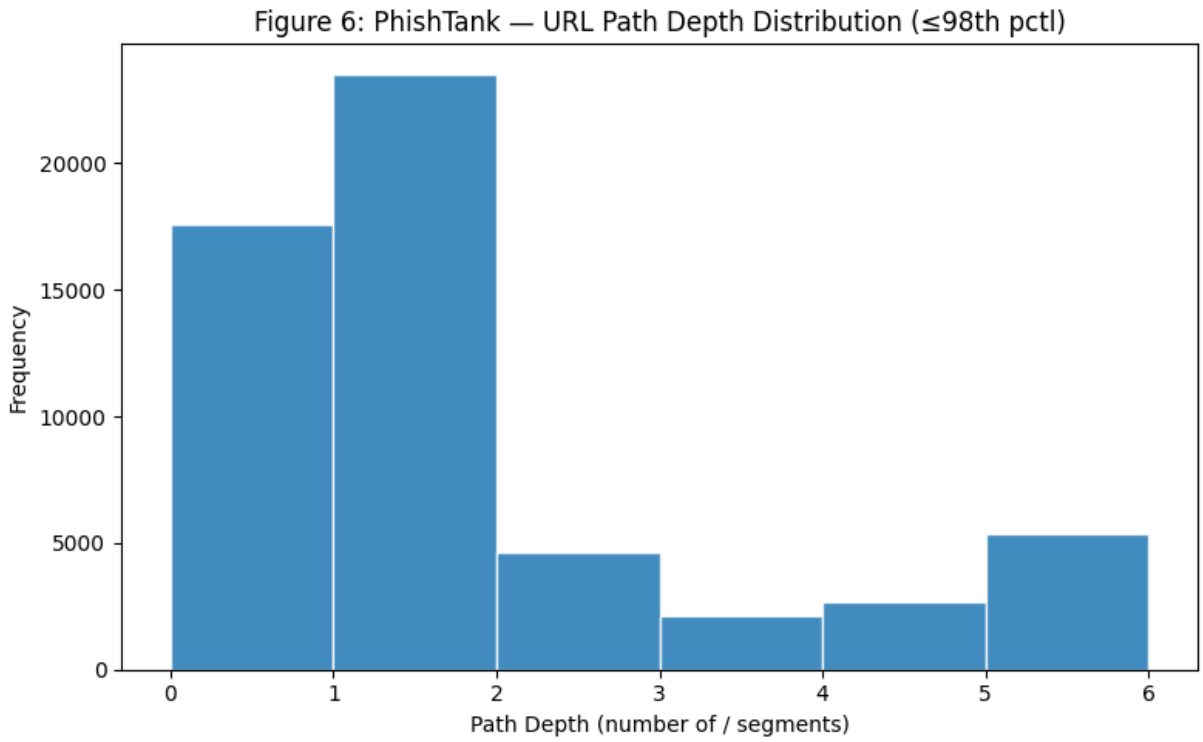
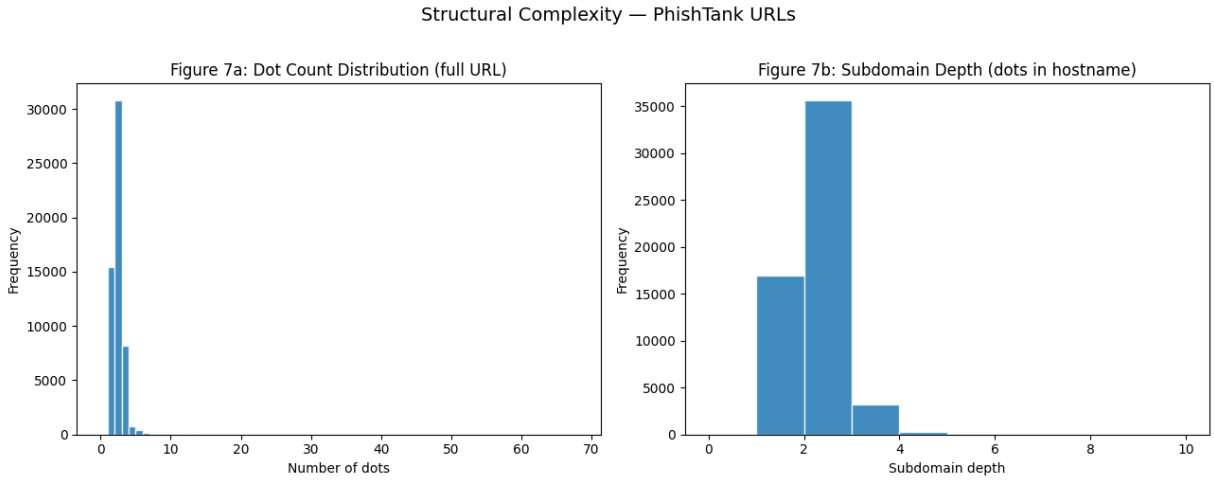


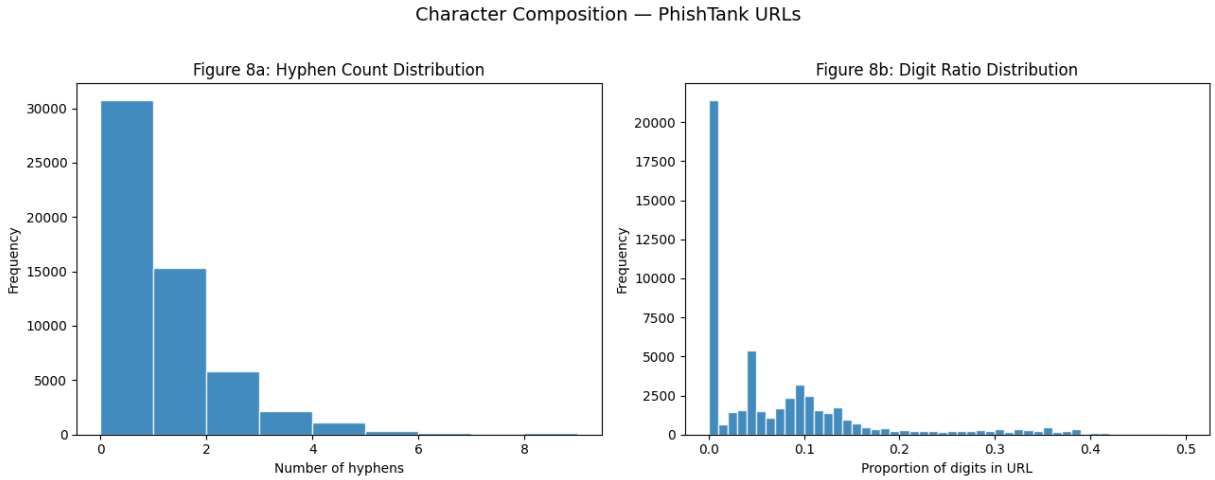
Figure 6 — Interpretation: Most phishing URLs have shallow path depths of 1-2 segments, though a notable portion have deeper paths. Deeper paths can indicate more sophisticated phishing kits with multiple redirect layers, which aligns with literature noting that complex URL structures are a common obfuscation technique. That said, there is not a strong sign that deeper or shallower paths are signs on their own of legitimacy or vice versa.

In [17]:
```
# Structural complexity: dot count & subdomain depth
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].hist(phish['dot_count'], bins=range(0, phish['dot_count'].max() + 2)
             edgecolor='white', alpha=0.85)
axes[0].set_title('Figure 7a: Dot Count Distribution (full URL)')
axes[0].set_xlabel('Number of dots')
axes[0].set_ylabel('Frequency')

axes[1].hist(phish['subdomain_depth'], bins=range(0, phish['subdomain_depth'
             edgecolor='white', alpha=0.85)
axes[1].set_title('Figure 7b: Subdomain Depth (dots in hostname)')
axes[1].set_xlabel('Subdomain depth')
axes[1].set_ylabel('Frequency')

fig.suptitle('Structural Complexity — PhishTank URLs', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



Figures 7a/7b — Interpretation: PhishTank URLs show higher dot counts than typical domains, reflecting deeper subdomain nesting. Subdomain depth follows a similar pattern, with many URLs using 2 or more subdomains, a common tactic to make malicious domains appear affiliated with legitimate ones (e.g. amazon.phish.com).

In [18]:
```
# Character composition: hyphens & digit ratio
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].hist(phish['hyphen_count'], bins=range(0, 10),
             edgecolor='white', alpha=0.85)
axes[0].set_title('Figure 8a: Hyphen Count Distribution')
axes[0].set_xlabel('Number of hyphens')
axes[0].set_ylabel('Frequency')

axes[1].hist(phish['digit_ratio'], bins=50,
              edgecolor='white', alpha=0.85, range=(0, 0.5))
axes[1].set_title('Figure 8b: Digit Ratio Distribution')
axes[1].set_xlabel('Proportion of digits in URL')
axes[1].set_ylabel('Frequency')

fig.suptitle('Character Composition — PhishTank URLs', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



Figures 8a/8b — Interpretation: Hyphens are rare but present in a notable subset of phishing URLs, sometimes used to mimic legitimate domains (e.g. paypal-secure.com). Digit ratios are low overall but with outliers — URLs with unusually high digit ratios may indicate algorithmically generated domains, which is a key signal for our planned ML analysis. We would additionally like to look into domains containing numbers masquerading as letters, e.g., '1' in place of an 'l', say "netf1ix.com".

Tranco — Top 1 M Legitimate Domains

This section analyzes the Tranco Top-1M ranking, an aggregated popularity list of legitimate domains.
Unlike PhishTank URLs, Tranco entries are bare domains (no scheme or path), making structural features directly comparable at the hostname level. Cisco provides resources to help massage all the data coming in [10].

List ID: 6GPNX | **Date accessed:** 03/05/2026

Data Dictionary — Tranco

Feature	Data Type	Description	Example
rank	int	Popularity rank (1 = most visited)	1
domain	string	Registered domain name	google.com
tld	string	Top-level domain extracted from domain	com
domain_length	int	Character length of the domain	10
dot_count	int	Number of dots in the domain	1
hyphen_count	int	Number of hyphens in the domain	0
digit_count	int	Number of digit characters in the domain	0
digit_ratio	float	Proportion of domain characters that are digits	0.0

In [19]:
```
tranco = pd.read_csv('../data/raw_input/top-1m.csv', header=None, names=['ra
tranco.head()
```

Out[19]:

	rank	domain
0	1	google.com
1	2	microsoft.com
2	3	gtld-servers.net
3	4	facebook.com
4	5	mail.ru

In [20]:
```
tranco.shape
```

Out[20]:
```
(1000000, 2)
```

In [21]:
```
tranco.dtypes
```

Out[21]:
```
rank      int64
domain      str
dtype: object
```

Null / Missing Value Analysis

In [22]:
```
tranco.isnull().sum()

(tranco == '').sum()
```

Out[22]:
```
rank      0
domain    0
dtype: int64
```

Duplicate Check

In [23]:
```
duplicateDomains = tranco['domain'].duplicated().sum()
duplicateRanks = tranco['rank'].duplicated().sum()
print(f'Duplicate domains: {duplicateDomains}')
print(f'Duplicate ranks: {duplicateRanks}')
```

```
Duplicate domains: 0
Duplicate ranks: 0
```

TLD Distribution

In [24]:
```
tranco['tld'] = tranco['domain'].str.split('.').str[-1]
tld_counts = tranco['tld'].value_counts().head(15)
print(tld_counts)
```

```
tld
com    431460
net     51741
ru      42879
org     39514
de      26817
uk      22815
br      17524
jp      16280
in      12403
fr      11433
io      10800
nl      10241
it      10051
xyz      9407
pl       9054
Name: count, dtype: int64
```

In [25]:
```
plt.figure(figsize=(12,6))
sns.barplot(x=tld_counts.values, y=tld_counts.index)
plt.title('Figure 9: Top 15 TLDs in Tranco Top 1M')
plt.xlabel('Count')
plt.ylabel('TLD')
plt.tight_layout()
plt.show()
```

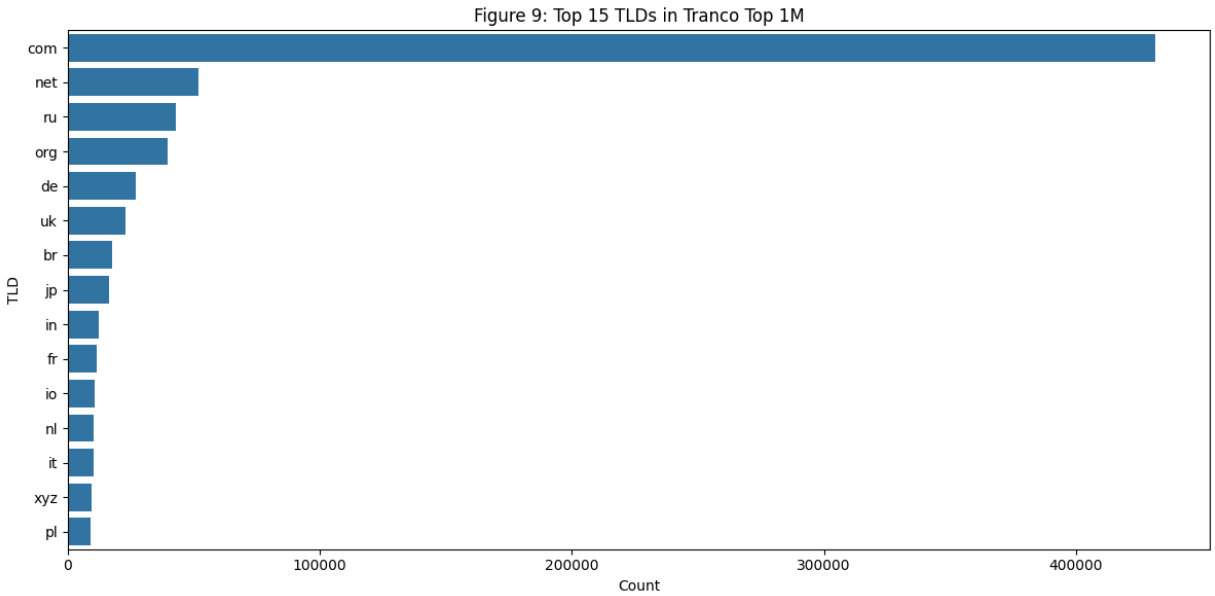


Figure 9 — Interpretation: .com dominates the Tranco top 1M as expected, reflecting its status as the default TLD for established businesses. Country-code TLDs like .ru, .de, and .br are also well-represented, indicating the dataset has strong international coverage which broadens the generalizability of our baseline.

Domain Length Distribution

In [26]:
```
tranco['domain_length'] = tranco['domain'].str.len()
print(tranco['domain_length'].describe())
```

```
count    1000000.000000
mean          13.739423
std            4.346079
min            4.000000
25%           11.000000
50%           13.000000
75%           16.000000
max           75.000000
Name: domain_length, dtype: float64
```

In [27]:
```
plt.figure(figsize=(10, 5))
sns.histplot(tranco['domain_length'], bins=50, kde=True)
plt.title('Figure 10: Domain Length Distribution (Tranco Top 1M)')
plt.xlabel('Character Length')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

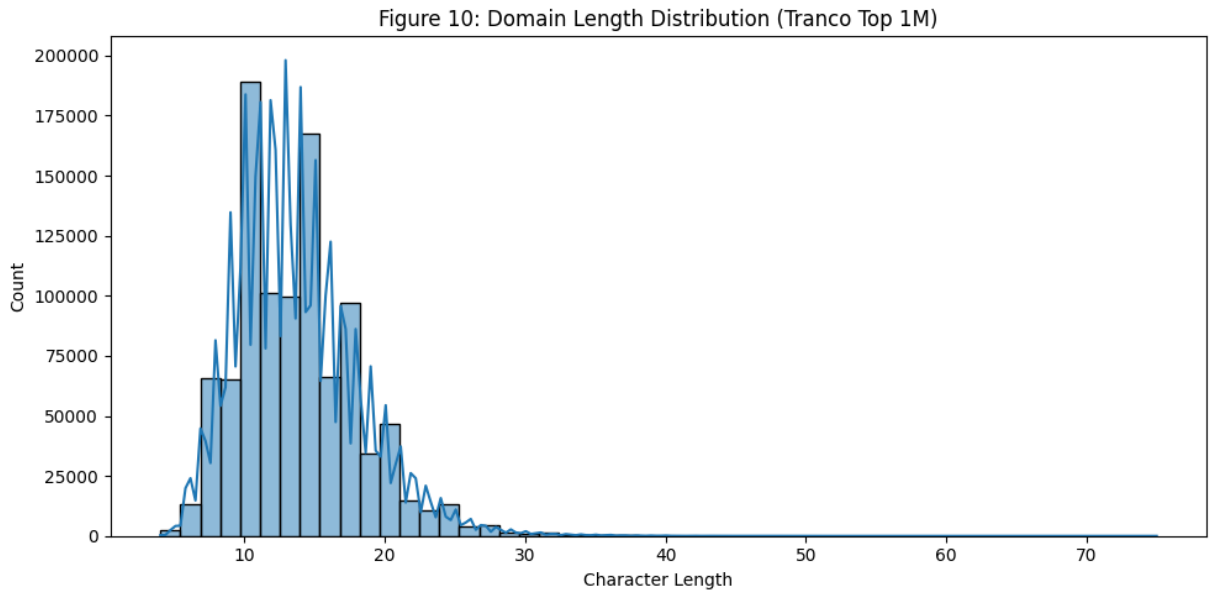


Figure 10 — Interpretation: Tranco domain lengths cluster tightly between 8-20 characters with a roughly normal distribution, reflecting the tendency of legitimate established domains to be short and memorable. This contrasts sharply with PhishTank URL lengths, providing a potentially strong discriminating feature for classification.

Structural & Character Features

Dot counts, subdomain depth, hyphens, and digit ratios for Tranco domains.

In [28]:
```
tranco['dot_count'] = tranco['domain'].str.count(r'\.')
tranco['hyphen_count'] = tranco['domain'].str.count('-')
tranco['digit_count'] = tranco['domain'].str.count(r'\d')
```

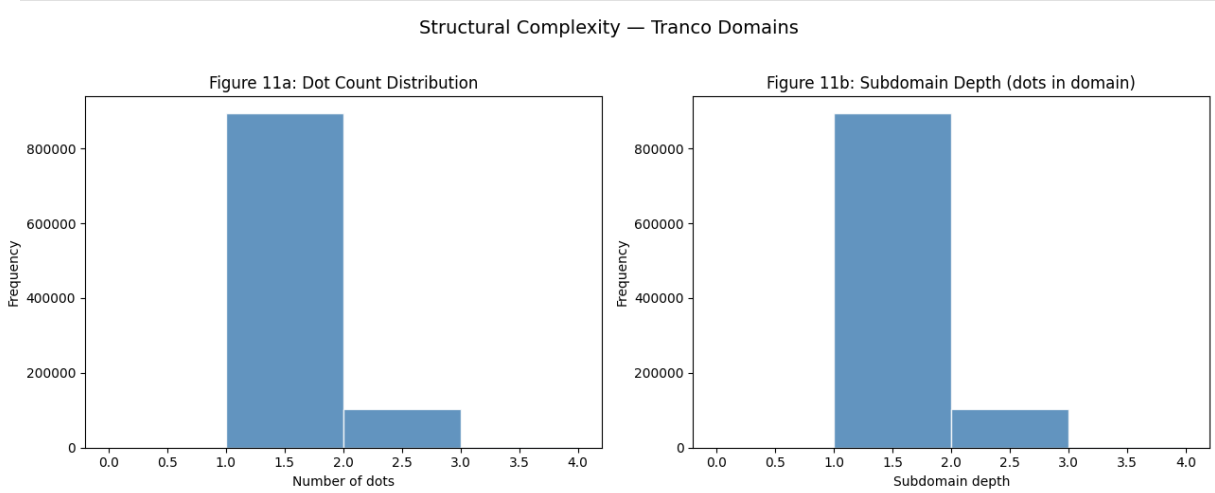
```
tranco['digit_ratio'] = tranco['digit_count'] / tranco['domain_length']
tranco['subdomain_depth'] = tranco['domain'].str.count(r'\.')

tranco[['domain_length', 'dot_count', 'hyphen_count', 'digit_count', 'digit_
```

Out[28]:

	domain_length	dot_count	hyphen_count	digit_count	dig
count	1000000.000000	1000000.000000	1000000.000000	1000000.000000	1000000.0
mean	13.739423	1.107173	0.121737	0.275257	0.0
std	4.346079	0.314686	0.410282	0.936712	0.0
min	4.000000	1.000000	0.000000	0.000000	0.0
25%	11.000000	1.000000	0.000000	0.000000	0.0
50%	13.000000	1.000000	0.000000	0.000000	0.0
75%	16.000000	1.000000	0.000000	0.000000	0.0
max	75.000000	3.000000	11.000000	56.000000	0.9

In [29]:

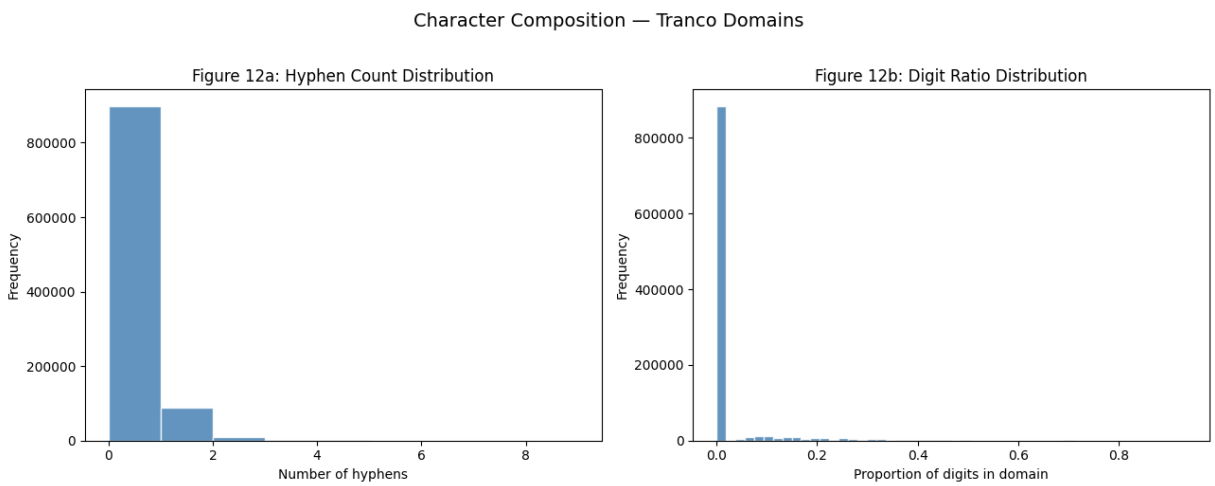
```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

bins_dots = range(0, tranco['dot_count'].max() + 2)
axes[0].hist(tranco['dot_count'], bins=bins_dots, color='steelblue', edgecol
axes[0].set_title('Figure 11a: Dot Count Distribution')
axes[0].set_xlabel('Number of dots')
axes[0].set_ylabel('Frequency')

bins_sub = range(0, tranco['subdomain_depth'].max() + 2)
axes[1].hist(tranco['subdomain_depth'], bins=bins_sub, color='steelblue', ed
axes[1].set_title('Figure 11b: Subdomain Depth (dots in domain)')
axes[1].set_xlabel('Subdomain depth')
axes[1].set_ylabel('Frequency')

fig.suptitle('Structural Complexity — Tranco Domains', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



Figures 11a/11b — Interpretation: Tranco domains predominantly have 1-2 dots,

corresponding to simple registered domains with at most one subdomain level. The low subdomain depth compared to PhishTank suggests subdomain nesting is a meaningful feature for distinguishing phishing from legitimate traffic.

In [30]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].hist(tranco['hyphen_count'], bins=range(0, 10), color='steelblue', e
axes[0].set_title('Figure 12a: Hyphen Count Distribution')
axes[0].set_xlabel('Number of hyphens')
axes[0].set_ylabel('Frequency')

axes[1].hist(tranco['digit_ratio'], bins=50, color='steelblue', edgecolor='w
axes[1].set_title('Figure 12b: Digit Ratio Distribution')
axes[1].set_xlabel('Proportion of digits in domain')
axes[1].set_ylabel('Frequency')

fig.suptitle('Character Composition — Tranco Domains', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



Figures 12a/12b — Interpretation: Tranco domains rarely contain hyphens and have very low digit ratios, reinforcing that legitimate popular domains tend to be clean alphanumeric strings. The contrast with PhishTank on both metrics supports their use as discriminating features in downstream ML modeling.

Cross-Dataset Analysis

PhishTank / Tranco Overlap

In [31]:
```
phish['domain'] = phish['url'].apply(lambda x:
    urlparse(x).netloc.lower().replace('www.', '', 1))

tranco_domains = set(tranco['domain'].str.lower())
```

```
overlap = phish['domain'].isin(tranco_domains).sum()
print(f'Phishing URLs whose domain appears in Tranco Top 1M: {overlap} / {le
```

```
Phishing URLs whose domain appears in Tranco Top 1M: 13662 / 56045 (24.38%)
```

In [32]:
```
print(f"Tranco: {len(tranco):,} legitimate domains (ranked by popularity)")
print(f"PhishTank: {len(phish):,} verified phishing URLs")
print(f"Overlap: {overlap:,} phishing domains found in Tranco top 1M ({overl
print("\nNotes:\n- Tranco aggregates multiple blocklists/popularity sources;
```

```
Tranco: 1,000,000 legitimate domains (ranked by popularity)
PhishTank: 56,045 verified phishing URLs
Overlap: 13,662 phishing domains found in Tranco top 1M (24.38%)

Notes:
- Tranco aggregates multiple blocklists/popularity sources; high rank != gua
ranteed safe
- PhishTank is a snapshot of verified-online phishing URLs, not exhaustive
- Overlap may reflect compromised legitimate domains or shared hosting infra
structure
```

Summary Comparison Table

Side-by-side comparison of Tranco legitimate domains vs PhishTank phishing URLs.

In [33]:
```
summary_data = {
    "Metric": [
        "Count", "Mean length", "Median length", "Max length",
        "Mean dots", "Mean hyphens", "Mean digit ratio",
        "Unique TLDs", "Mean subdomain depth"
    ],
    "Tranco (domain)": [
        f"{len(tranco):,}",
        f"{tranco['domain_length'].mean():.1f}",
        f"{tranco['domain_length'].median():.0f}",
        f"{tranco['domain_length'].max()}",
        f"{tranco['dot_count'].mean():.2f}",
        f"{tranco['hyphen_count'].mean():.2f}",
        f"{tranco['digit_ratio'].mean():.4f}",
        f"{tranco['tld'].nunique()}",
        f"{tranco['subdomain_depth'].mean():.2f}",
    ],
    "PhishTank (full URL)": [
        f"{len(phish):,}",
        f"{phish['full_length'].mean():.1f}",
        f"{phish['full_length'].median():.0f}",
        f"{phish['full_length'].max()}",
        f"{phish['dot_count'].mean():.2f}",
        f"{phish['hyphen_count'].mean():.2f}",
        f"{phish['digit_ratio'].mean():.4f}",
        f"{phish['tld_ext'].nunique()}",
        f"{phish['subdomain_depth'].mean():.2f}",
    ],
}
summary_tbl = pd.DataFrame(summary_data)
```

```
fig, ax = plt.subplots(figsize=(10, 4))
ax.axis("off")
tbl = ax.table(cellText=summary_tbl.values, colLabels=summary_tbl.columns,
               loc="center", cellLoc="center")
tbl.auto_set_font_size(False)
tbl.set_fontsize(10)
tbl.scale(1.2, 1.6)
for (row, col), cell in tbl.get_celld().items(): # From Datacamp [9], altern
    if row == 0:
        cell.set_facecolor("#4472C4")
        cell.set_text_props(color="white", fontweight="bold")
    elif row % 2 == 0:
        cell.set_facecolor("#D9E2F3")
ax.set_title("Figure 12: Side-by-Side Comparison: Tranco vs PhishTank", font
             fontweight="bold", pad=20)
plt.tight_layout()
plt.show()
```

Figure 12: Side-by-Side Comparison: Tranco vs PhishTank

Metric	Tranco (domain)	PhishTank (full URL)
Count	1,000,000	56,045
Mean length	13.7	60.9
Median length	13	38
Max length	75	25523
Mean dots	1.11	1.97
Mean hyphens	0.12	0.80
Mean digit ratio	0.0225	0.0750
Unique TLDs	1042	285
Mean subdomain depth	1.11	1.77

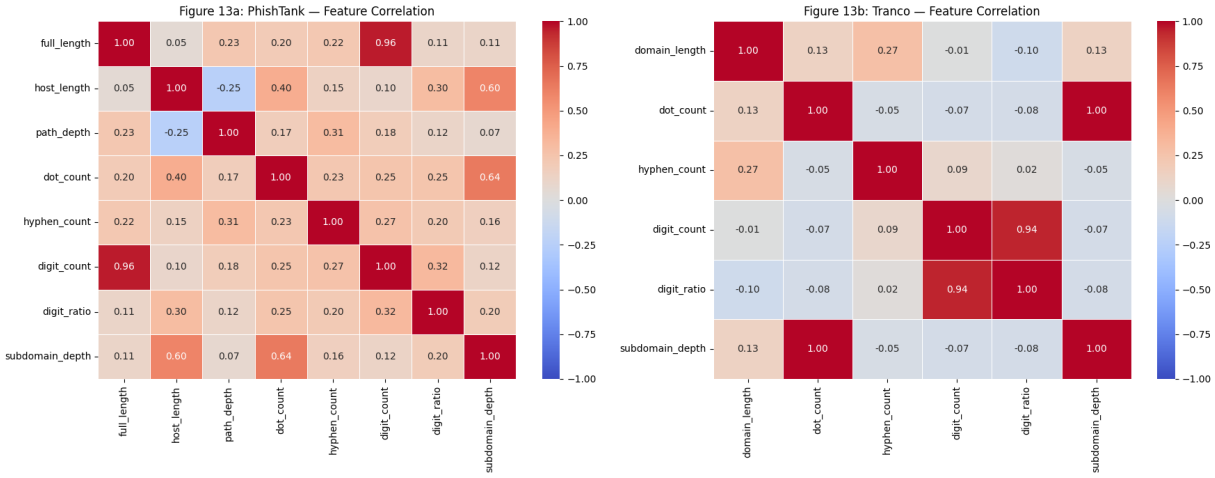
Figure 12 — Interpretation: Side-by-side comparison shows systematic structural differences between PhishTank and Tranco. PhishTank URLs are longer, have higher hyphen counts, and greater subdomain depth on average. These differences are consistent across the dataset and suggest that lexical URL features alone carry meaningful signal for phishing detection.

Correlation Heatmaps

Pearson correlation among the numerical structural features for each dataset. Since Tranco entries are bare domains and PhishTank entries are full URLs, each heatmap uses the features native to its dataset.

In [34]:
```
phish_num = phish[['full_length', 'host_length', 'path_depth', 'dot_count',
                   'hyphen_count', 'digit_count', 'digit_ratio', 'subdomain_
tranco_num = tranco[['domain_length', 'dot_count', 'hyphen_count',
                     'digit_count', 'digit_ratio', 'subdomain_depth']]

fig, axes = plt.subplots(1, 2, figsize=(18, 7))
```

```
# PhishTank heatmap
corr_phish = phish_num.corr()
sns.heatmap(corr_phish, annot=True, fmt='.2f', cmap='coolwarm', center=0,
            linewidths=0.5, ax=axes[0], vmin=-1, vmax=1)
axes[0].set_title('Figure 13a: PhishTank — Feature Correlation', fontsize=12

# Tranco heatmap
corr_tranco = tranco_num.corr()
sns.heatmap(corr_tranco, annot=True, fmt='.2f', cmap='coolwarm', center=0,
            linewidths=0.5, ax=axes[1], vmin=-1, vmax=1)
axes[1].set_title('Figure 13b: Tranco — Feature Correlation', fontsize=12)

plt.tight_layout()
plt.show()

# Interpret strongest correlations for each dataset
for name, corr in [('PhishTank', corr_phish), ('Tranco', corr_tranco)]:
    # Upper triangle only (no duplicates / diagonal)
    mask = pd.np.triu(pd.np.ones(corr.shape), k=1).astype(bool) if hasattr(p
    pairs = corr.where(mask).stack().reset_index()
    pairs.columns = ['Feature A', 'Feature B', 'r']
    pairs['abs_r'] = pairs['r'].abs()
    top = pairs.nlargest(3, 'abs_r')
    print(f"\n{name} — Top 3 strongest correlations:")
    for _, row in top.iterrows():
        print(f"  {row['Feature A']} ↔ {row['Feature B']}: r = {row['r']:.3f
```




```
PhishTank — Top 3 strongest correlations:
  full_length ↔ digit_count: r = 0.957
  dot_count ↔ subdomain_depth: r = 0.643
  host_length ↔ subdomain_depth: r = 0.599

Tranco — Top 3 strongest correlations:
  dot_count ↔ subdomain_depth: r = 1.000
  digit_count ↔ digit_ratio: r = 0.945
  domain_length ↔ hyphen_count: r = 0.274
```

Figures 13a/13b — Interpretation: The PhishTank correlation heatmap reveals strong positive correlation between full URL length and host length, as well as dot count and subdomain depth — as expected since longer URLs tend to have more components.

Tranco shows weaker inter-feature correlations overall, suggesting its domains are structurally more uniform and simple.

Scatter Plots — Feature Relationships

Pair plots of key structural features for each dataset to visualize bivariate relationships.

In [35]:
```
import numpy as np

# Build a combined dataframe with a 'source' label as a target variable
phish_sample = phish[['host_length', 'dot_count', 'hyphen_count', 'digit_rat
phish_sample['source'] = 'PhishTank'

tranco_sample = tranco[['domain_length', 'dot_count', 'hyphen_count', 'digit
tranco_sample = tranco_sample.rename(columns={'domain_length': 'host_length'
tranco_sample['source'] = 'Tranco'

combined = pd.concat([phish_sample, tranco_sample], ignore_index=True)

# Pair plot colored by dataset source
g = sns.pairplot(combined.sample(n=min(4000, len(combined)), random_state=42
                 hue='source', vars=['host_length', 'dot_count', 'hyphen_cou
                 diag_kind='hist', plot_kws={'alpha': 0.4, 's': 15},
                 palette={'PhishTank': 'salmon', 'Tranco': 'steelblue'})
g.figure.suptitle('Figure 14: Pair Plot — PhishTank vs Tranco Structural Fea
plt.tight_layout()
plt.show()
```

Figure 14: Pair Plot — PhishTank vs Tranco Structural Features

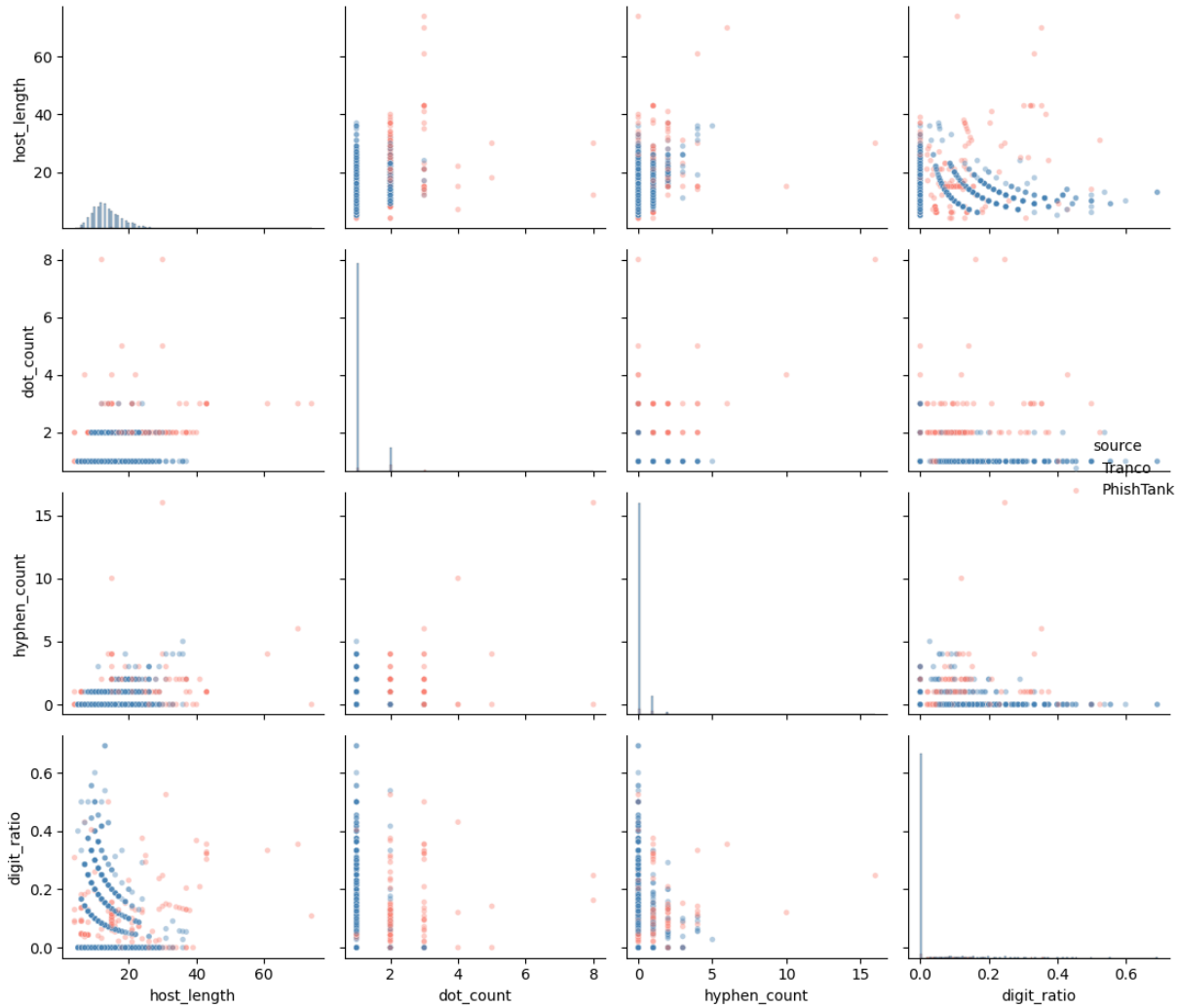


Figure 14 — Interpretation: The pair plot contrasts PhishTank and Tranco samples across key structural features. Clear separation is visible in several feature pairs, particularly URL length and subdomain depth, indicating that a linear or tree-based classifier should be able to leverage these features effectively for binary phishing detection.

Data Normalization / Scaling

StandardScaler applied for future use for ML purposes.

In [36]:
```
from sklearn.preprocessing import StandardScaler

scale_features = ['full_length', 'host_length', 'path_depth', 'dot_count',
                  'hyphen_count', 'digit_ratio', 'subdomain_depth']

scaler = StandardScaler()
phish_scaled = pd.DataFrame(scaler.fit_transform(phish[scale_features]),
                            columns=[f'{c}_scaled' for c in
                                     scale_features])
```

```
phish_scaled.describe().round(2)
```

Out[36]:

	full_length_scaled	host_length_scaled	path_depth_scaled	dot_count_scaled	hy
count	56045.00	56045.00	56045.00	56045.00	
mean	0.00	-0.00	0.00	0.00	
std	1.00	1.00	1.00	1.00	
min	-0.29	-1.18	-0.88	-0.95	
25%	-0.23	-0.87	-0.88	-0.95	
50%	-0.14	-0.25	-0.24	0.02	
75%	0.01	0.44	0.39	0.02	
max	153.03	9.84	10.50	63.41	

Export Processed Datasets

In [37]:

```
phish.to_csv('../data/processed_input/phishtank_cleaned.csv', index=False)
tranco.to_csv('../data/processed_input/tranco_cleaned.csv', index=False)
print("Exported phishtank_cleaned.csv and tranco_cleaned.csv.")
```

```
Exported phishtank_cleaned.csv and tranco_cleaned.csv.
```

Success.

Citations

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